

● HARD

● INFORMATION AND IDEAS

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05

10 questions · ~15 min

Q1

INFORMATION AND IDEAS

Birds of many species ingest foods containing carotenoids, pigmented molecules that are converted into feather coloration. Coloration tends to be especially saturated in male birds' feathers, and because carotenoids also confer health benefits, the deeply saturated colors generally serve to communicate what is known as an honest signal of a bird's overall fitness to potential mates. However, ornithologist Allison J. Shultz and others have found that males in several species of the tanager genus *Ramphocelus* use microstructures in their feathers to manipulate light, creating the appearance of deeper saturation without the birds necessarily having to maintain a carotenoid-rich diet. These findings suggest that _____blank

Which choice most logically completes the text?

- A. individual male tanagers can engage in honest signaling without relying on carotenoid consumption.
- B. feather microstructures may be less effective than deeply saturated feathers for signaling overall fitness.
- C. scientists have yet to determine why tanagers have a preference for mates with colorful appearances.
- D. a male tanager's appearance may function as a dishonest signal of the individual's overall fitness.

Electronic music pioneer Wendy Carlos is credited with the music for three feature films: *A Clockwork Orange* (1971), *The Shining* (1980), and *Tron* (1982). However, her musical score for *A Clockwork Orange* is mostly made up of her arrangements of Ludwig van Beethoven's work. Also, almost all the music that she and Rachel Elkind composed for *The Shining* was unused by director Stanley Kubrick. It did not appear in the film. Of the three films, *Tron* is the one in which audiences can hear the most of Carlos's original compositions.

Which choice best states the main idea of the text?

- A. Of the three films for which Carlos is credited, *Tron* features the most original music from her.
- B. The director of *The Shining* used most of the music that Carlos composed for it.
- C. Beethoven is widely considered to be a more important composer than Carlos.
- D. Carlos is a notable innovator among film composers in the 1970s and 1980s.

Electra is a circa 420–410 BCE play by Sophocles, translated in 1870 by R.C. Jebb. Electra, who is in mourning for her dead father and her long-absent brother, is aware of the intensity of her grief but believes it to be justified: _____ blank

Which quotation from *Electra* most effectively illustrates the claim?

- A. “O thou pure sunlight, and thou air, earth’s canopy, how often have ye heard the strains of my lament, the wild blows dealt against this bleeding breast, when dark night fails!”
- B. “Send to me my brother; for I have no more the strength to bear up alone against the load of grief that weighs me down.”
- C. “I know my own passion, it escapes me not; but, seeing that the causes are so dire, will never curb these frenzied complaints, while life is in me.”
- D. “But never will I cease from dirge and sore lament, while I look on the trembling rays of the bright stars, or on this light of day.”

Many animals, including humans, must sleep, and sleep is known to have a role in everything from healing injuries to encoding information in long-term memory. But some scientists claim that, from an evolutionary standpoint, deep sleep for hours at a time leaves an animal so vulnerable that the known benefits of sleeping seem insufficient to explain why it became so widespread in the animal kingdom. These scientists therefore imply that _____ blank

Which choice most logically completes the text?

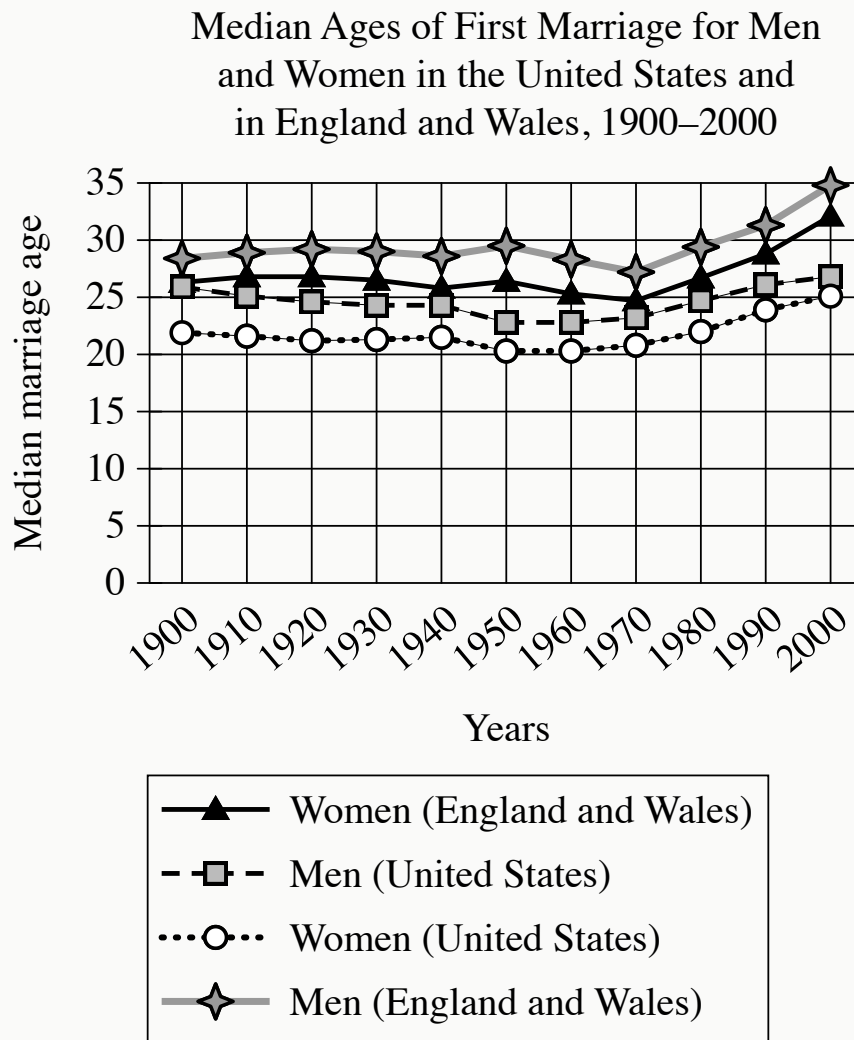
- A. prolonged deep sleep is likely advantageous in ways that have yet to be discovered.
- B. most traits perform functions that are hard to understand from an evolutionary standpoint.
- C. it is more important to understand how widespread prolonged deep sleep is than to understand its function.
- D. many traits that provide significant benefits for an animal also likely pose risks to that animal.

The following text is adapted from Countee Cullen's 1926 poem "Thoughts in a Zoo."

They in their cruel traps, and we in ours,
Survey each other's rage, and pass the hours
Commiserating each the other's woe,
To mitigate his own pain's fiery glow.
Man could but little proffer in exchange
Save that his cages have a larger range.
That lion with his lordly, untamed heart
Has in some man his human counterpart,
Some lofty soul in dreams and visions wrapped,
But in the stifling flesh securely trapped.

Based on the text, what challenge do humans sometimes experience?

- A. They cannot effectively tame certain wild animals because of a lack of compassion.
- B. They cannot focus on setting attainable goals because of a lack of motivation.
- C. They quickly become frustrated when faced with difficult tasks because of a lack of self-control.
- D. They have aspirations that cannot be fulfilled because of certain limitations.



A sociology student is reading an essay on the median age of first marriage in Western countries throughout the twentieth century. The author of the essay cites factors common to these countries that the author believes caused an increase in the median age of first marriage, such as new technologies that shortened the time needed for domestic chores, making two-person households less necessary and living alone more viable. The student asserts that beyond these factors there must be additional ones specific to particular Western countries that influenced the increase of age at first marriage.

Which choice most effectively uses data from the graph that support the student's assertion?

- A. Between 1970 and 2000, the median age of first marriage rose more sharply for men in England and Wales than it did for men in the United States.
- B. In England and Wales, the median age of first marriage was consistently higher for men than for women between 1900 and 2000, but this was not always the case in the United States.
- C. The median age of first marriage for men in England and Wales was lower in 1970 than in 1950 or 1990.
- D. Between 1900 and 2000, the median age of first marriage for women in England and Wales was consistently higher than for women in the United States, as was the case for men.

Tides can deposit large quantities of dead vegetation within a salt marsh, smothering healthy plants and leaving a salt panne—a depression devoid of plants that tends to trap standing water—in the marsh’s interior. Ecologist Kathryn Beheshti and colleagues found that burrowing crabs living within these pannes improve drainage by loosening the soil, leading the pannes to shrink as marsh plants move back in. At salt marsh edges, however, crab-induced soil loosening can promote marsh loss by accelerating erosion, suggesting that the burrowing action of crabs _____ blank

Which choice most logically completes the text?

- A. can be beneficial to marshes with small pannes but can be harmful to marshes with large pannes.
- B. may promote increases in marsh plants or decreases in marsh plants, depending on the crabs’ location.
- C. tends to be more heavily concentrated in areas of marsh interiors with standing water than at marsh edges.
- D. varies in intensity depending on the size of the panne relative to the size of the surrounding marsh.

Artificial leaves are a developing renewable energy technology that mimics the process of photosynthesis in plants. These devices are silicon-based solar cells coated in chemical catalysts that activate reactions that split water molecules into hydrogen and oxygen gas. The technology, while generating lots of interest, is not yet commercially viable as a large-scale energy source. To meet this challenge, scientists from many fields are researching ways to store, transport, and distribute the energy the devices produce while other scientists are working to improve the cost and efficiency of the devices.

Which choice best states the main idea of the text?

- A. Continued research and development in artificial-leaf technology is needed before the devices can be widely used as an energy source.
- B. The recent increase in the commercial use of artificial leaves as an energy source has encouraged many scientists to research ways to improve the technology.
- C. Artificial leaves split water molecules into oxygen and hydrogen gas using catalysts more efficiently than plants do using the process of photosynthesis.
- D. Artificial leaves were developed to mimic the natural process of photosynthesis in plants in order to store energy for long-term commercial use.

Number and Origin of Clamshell Tools Found at Different Levels Below the Surface in Neanderthal Cave

Depth of tools found below surface in cave (meters)	Clamshells that Neanderthals collected from the beach	Clamshells that Neanderthals harvested from the seafloor
3–4	99	33
6–7	1	0
4–5	2	0
2–3	7	0
5–6	18	7

Studying tools unearthed at a cave site on the western coast of Italy, archaeologist Paola Villa and colleagues have determined that prehistoric Neanderthal groups fashioned them from shells of clams that they harvested from the seafloor while wading or diving or that washed up on the beach. Clamshells become thin and eroded as they wash up on the beach, while those on the seafloor are smooth and sturdy, so the research team suspects that Neanderthals prized the tools made with seafloor shells. However, the team also concluded that those tools were likely more challenging to obtain, noting that _____blank

Which choice most effectively uses data from the table to support the research team's conclusion?

- A. at each depth below the surface in the cave, the difference in the numbers of tools of each type suggests that shells were easier to collect from the beach than to harvest from the seafloor.

- B.** the highest number of tools were at a depth of 3–4 meters below the surface, which suggests that the Neanderthal population at the site was highest during the related period of time.
- C.** at each depth below the surface in the cave, the difference in the numbers of tools of each type suggests that Neanderthals preferred to use clamshells from the beach because of their durability.
- D.** the higher number of tools at depths of 5–6 meters below the surface in the cave than at depths of 4–5 meters below the surface suggests that the size of clam populations changed over time.

In the early nineteenth century, some Euro-American farmers in the northeastern United States used agricultural techniques developed by the Haudenosaunee (Iroquois) people centuries earlier, but it seems that few of those farmers had actually seen Haudenosaunee farms firsthand. Barring the possibility of several farmers of the same era independently developing techniques that the Haudenosaunee people had already invented, these facts most strongly suggest that _____ blank

Which choice most logically completes the text?

- A. those farmers learned the techniques from other people who were more directly influenced by Haudenosaunee practices.
- B. the crops typically cultivated by Euro-American farmers in the northeastern United States were not well suited to Haudenosaunee farming techniques.
- C. Haudenosaunee farming techniques were widely used in regions outside the northeastern United States.
- D. Euro-American farmers only began to recognize the benefits of Haudenosaunee farming techniques late in the nineteenth century.